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A Systematic Review of Augmented Reality Applications for Automated Driving: 2009–2020

Abstract

While augmented reality (AR) interfaces have been researched extensively over the last decades, studies on their application in vehicles have only recently advanced. In this paper, we systematically review 12 years of AR research in the context of automated driving (AD), from 2009 to 2020. Due to the multitude of possibilities for studies with regard to AR technology, at present, the pool of findings is heterogeneous and non-transparent. From a review of the literature we identified $N = 156$ papers with the goal to analyze the status quo of existing AR studies in AD, and to classify the related literature into application areas. We provide insights into the utilization of AR technology used at different levels of vehicle automation, and for different users (drivers, passengers, pedestrians) and tasks. Results show that most studies focused on safety aspects, driving assistance, and designing non-driving-related tasks. AR navigation, trust in automated vehicles (AVs), and interaction experiences also marked a significant portion of the published papers; however, a wide range of different parameters was investigated by researchers. Among other things, we find that there is a growing trend toward simulating AR content within virtual driving simulators. We conclude with a discussion of open challenges, and give recommendations for future research in automated driving at the AR side of the reality-virtuality continuum.

I Introduction

The development of automated driving (AD) systems and the accompanying human-machine interfaces (HMI) currently represent one of the biggest changes in mobility and transportation research and development. In 2021, the Society of Automotive Engineers (SAE) (SAE On-Road Automated Vehicle Standards Committee, 2021) published the latest version of its definition of vehicle automation and its various levels to address challenges and establish the basis for future standardization, as well as a common language. With SAE Level 2 (L2) partially automated vehicles being already on the road, it is only a matter of time before SAE Level 3 (L3) or even higher levels for automated driving are commercially available. Especially in the automotive sector, advances in automated driving and augmented reality technology are pushing academia and industry to develop new concepts for in-vehicle user interfaces. A growing number of HMI concepts in information visualization along the

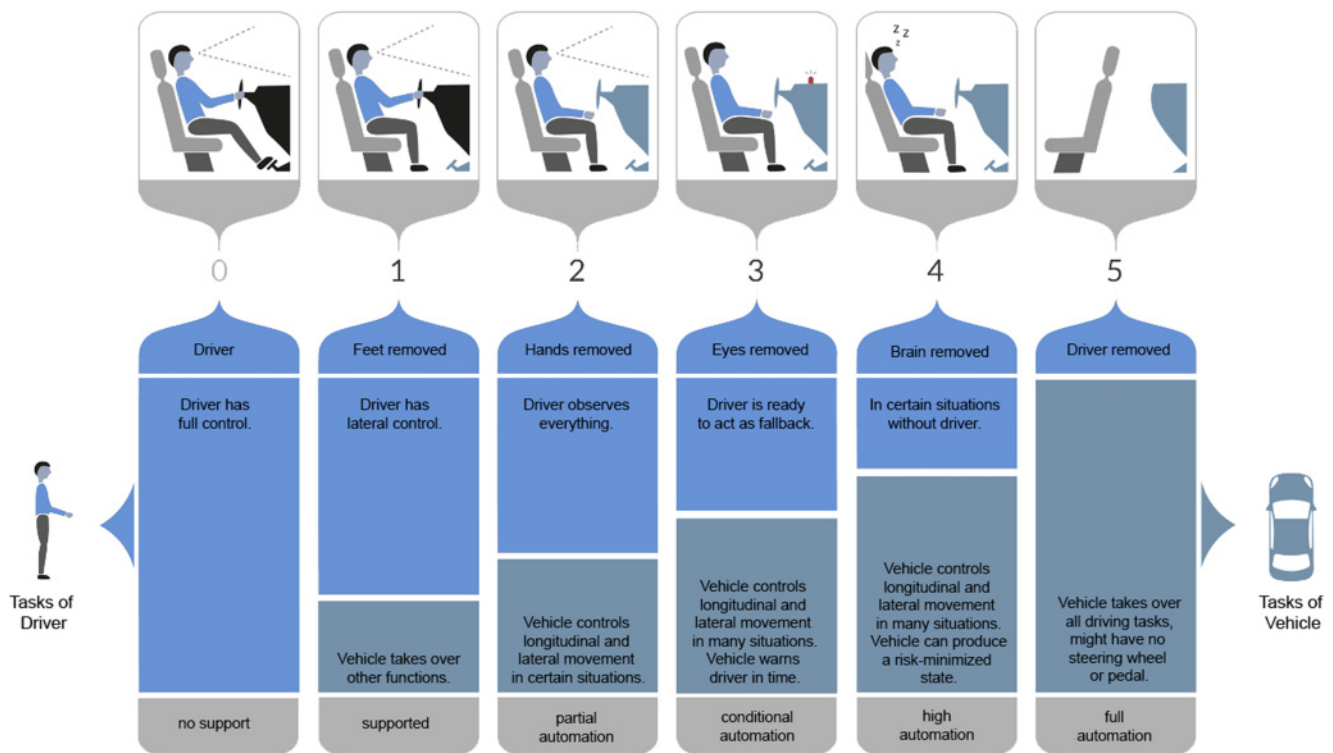


Figure 1. SAE levels of vehicle automation (SAE On-Road Automated Vehicle Standards Committee, 2021).

reality-virtuality (RV) continuum (Milgram, Takemura, Utsumi, & Kishino, 1995), from traditional 2D screens to augmented reality (AR) and virtual reality (VR), are being explored. Application areas include automotive, aviation, logistics, manufacturing, and education, among others (Dey, Billinghurst, Lindeman, & Swan, 2018). In AR, digital virtual images are overlaid on the real world so that the digital content is aligned with real-world objects (Azuma, 1997). In contrast, in VR, environments consist exclusively of 3D digital objects (Steuer, 1992). Consequently, mixed reality (MR) encompasses all possible positions on the RV continuum, that is, between the real world and the virtual environment. For the automotive sector in particular, MR technology has been used for a wide range of use cases. For example, in prototype user studies, head-up displays (HUDs) and larger windshield displays (WSDs) use AR technology to increase driver situational awareness by keeping the driver's eyes on the road, and go even further by presenting work- or entertainment-related content for performing non-

driving related tasks (NDRTs) in highly automated vehicles.

1.1 The Role of AR in Driving Research

The evolution of vehicle automation from manual and assisted driving to highly and fully automated vehicles necessitates a categorization for these different automation levels, as mixed traffic where manual and automated vehicles share the road is expected to become more frequent. Therefore, the SAE On-Road Automated Vehicle Standards Committee (2021) has introduced a common language for defining these levels of automation (see Figure 1). Each level of vehicle automation presents both opportunities and challenges for researchers and practitioners. While lower levels require consideration of situational awareness and safety concerns, higher levels of automation could bring new interaction and passenger experiences to the vehicle.

AR systems for vehicles, such as video-based AR displays that assist drivers with steering and parking, are already in use. Gabbard, Fitch, and Kim (2014) introduced a conceptual design space for automotive AR applications that considers display devices such as head-up displays, head-mounted displays, and device displays, as well as information presentation modality such as screen-fixed and world-fixed content presentations. In screen-fixed displays, digital content is superimposed on a fixed location on the device screen, while in world-fixed AR, graphics are rendered to appear to be in the external environment (Gabbard et al., 2014). With the advancement of AR and eye-tracking technology, screen-fixed head-up displays are increasingly transitioning to world-fixed AR windshield displays (Blankenbach, 2018). Furthermore, mixed reality (in general) can be categorized by the level of immersion provided to users. Non-immersive visualizations, such as desktop monitor setups, are easy and inexpensive to implement. In particular, previous studies of head-up displays have been conducted with 2D monitors that simulate AR functions in a non-immersive manner. Semi-immersive mixed reality is achieved with head-up displays (HUDs), which provide an extension of the real world through a digital overlay. In contrast, head-mounted displays (HMDs) and Cave Automatic Virtual Environments (CAVEs) (Cruz-Neira, Sandin, DeFanti, Kenyon, & Hart, 1992), which consist of projections on walls, provide a high level of immersion for the user as the real environment fades into the background.

VR technology has been applied to explore automated driving functions and AR experiences in safe and controlled environments. Studies indicate that participants' subjective and psychological responses in virtual reality environments are closely tied to their experiences and behaviors in a real-world environment (Slater et al., 2006). Moreover, VR configurations enable controlled, reproducible, and safe experiments (De Winter, van Leeuwen, & Happee, 2012). In addition, data collection (e.g., driving behavior) is less tedious than for real vehicles (De Winter et al., 2012). Pettersson, Karlsson, and Ghiurau (2019) compared a VR setup to a field study and found that while VR studies cannot completely be

a replacement for field studies, they can add value to driving simulators and desktop testing. Another comparative study conducted by Langlois, That, and Mermillod (2016) found that the virtual environment is able to replicate the relative distance perception between virtual objects (e.g., vehicles, pedestrians, road objects) and their overlays. Therefore, Langlois et al. (2016) recommend that AR HUDs or WSDs can be simulated in user experiments in virtual environments. Other comparative studies, including Bolder, Grünvogel, and Angelescu (2018), Rhiu, Kim, Kim, and Yun (2020), and Weidner, Heesch, Poeschl, and Broll (2017), also demonstrate the feasibility and practicality of using VR for prototyping and investigating automotive AR experiences and applications.

There is a broad range of challenges that need to be addressed, and a large number of papers have been published in recent years that address these topical issues. However, it is often difficult to integrate and/or compare the results that have been obtained because the topics are often studied in different ways. In addition, the sheer volume of findings makes it difficult for researchers to identify gaps on which to build. Therefore, we believe that it is time to systematically review which topics have received the most attention in research on augmented reality applications in driving automation and what methods have been used to address these topics. To help the automotive user interface (UI) community improve the usability and user experience of AR applications, this paper provides an overview of 12 years of AR studies conducted in the automotive domain. We chose to start with 2009, as this was the year that the AutomotiveUI community (AutomotiveUI, 2009) was launched, with a particular emphasis on user interface design and HCI research in the automotive context. The overarching goal of this literature review is to provide an overview of current and past adoption of AR in automated vehicles, in terms of their level of automation, to help researchers and practitioners find examples of related studies, and to help identify research gaps that should be addressed by future studies. Thus, this survey reflects to the automotive UI community on what has already been achieved and which future issues require more consideration.

1.2 Topics of Interest

The principal scope of this survey focuses on the combination of two emerging technologies: (1) augmented reality applications and usage, and (2) automated driving systems. Gartner's hype cycle (Gartner, Inc., 2021), an indicator of common trends that arise in emerging technologies, illustrates the current maturity of these technologies, and is indicative of the growth or decline of a particular technology. In 2014, both augmented and virtual reality were in the "Trough of Disillusionment" phase, meaning that these technologies were being subjected to rigorous testing with experiments and prototype deployments (Gartner, Inc., 2014). Five years later, both AR and VR have been removed from the Gartner hype cycle for emerging technologies (Gartner, Inc., 2019), indicating that they have left experimental status and have become mature, that is, usable and useful. SAE level 4 autonomous driving has not yet reached this status. For automotive HCI researchers and practitioners, this means that MR will find its way into conditionally automated vehicles (AVs) in the foreseeable future. Consequently, according to Gabbard et al. (2014), manufacturers are anticipated to build on these technologies and offer AR applications in the near term. In the meantime, VR technology is being utilized to simulate AR applications because technical and financial constraints have not yet allowed widespread implementation and deployment of dedicated AR systems (Gerber, Schroeter, & Vehns, 2019; Riegler, Riener, & Holzmann, 2019b). We aim, therefore, to revisit the role played by **augmentation** and **virtuality** and inherently explore the AR side of Milgram's RV continuum (Milgram et al., 1995).

We also want to consider that AR technology could be viewed as a "means to an end," that is, we intend to explore how this technology can be used to calibrate the trust and acceptance of automated driving. Thus, another major topic we want to address are **application areas**. We are interested in the developments from early AR applications for non- and partially automated vehicles (up to SAE L2) to new concepts and studies for fully automated vehicles (SAE L5). AR application research thus aims to explore the underlying dynamics required

to replace previous driver experiences with other potentially meaningful non-driving related tasks (NDRTs), as well as appropriate user interfaces that highlight the benefits and offset the negative effects of automation, further advancing vehicles into mobile work and living spaces (Schartmüller, Riener, Wintersberger, & Frison, 2018).

As automated driving systems become more advanced, traditional drivers will become passive passengers. Therefore, another focus of this survey is on the (potential) **users** of AR technology in and around vehicles, including drivers, passengers, pedestrians, and other road users. Since MR technology can be used to explore relevant scenarios, for example, using simulated environments, we give an overview of the **devices** used for augmented reality research related to automated driving. These include head-mounted displays (HMDs), HUDs, WSDs, Cave Automatic Virtual Environments (CAVEs), 2D monitor setups, and projections.

1.3 Previous Literature Reviews and User Study Survey Papers

Dey et al. (2018) provide a systematic review of augmented reality usability studies, which among other application areas, includes automotive applications. Contrarily, our systematic review focuses completely on automotive AR with additional emphasis on vehicle automation. In contrast to the recent survey on user studies on automated driving conducted by Frison, Forster, Wintersberger, Geisel, and Riener (2020), this work reviews and discusses user studies on vehicles with different levels of driving automation conducted for automotive AR applications. In 2020, Riegler, Riener, and Holzmann (2020b) presented workshop results on challenges and opportunities in mixed reality applications for future mobility, and therefore outlines a research roadmap, whereas our view is more focused on AR as a subtopic of mixed reality, and covers both the main research areas of recent user studies, and how these studies were carried out. Gabbard et al. (2014) focus on driver challenges and opportunities for in-vehicle AR applications. We focus on analyzing methodological approaches and application areas from empirical user studies run

with AR technology in vehicles, and derive suggestions and action points for future studies in this field. McGill and Brewster (2019), in their review, look at user studies related to passenger experiences with HMDs, while we take a holistic approach towards AR usage for driving-related user studies, including passenger experiences. Furthermore, Rouchitsas and Alm (2019) provide a review of external human-machine interface (eHMI) research. While their focus was on empirical user studies on eHMIs in general, we explore user studies using AR HMIs utilized by pedestrians as well as other road users. The role of VR in AD research was investigated by Riegler, Riener, and Holzmann (2021). In the present literature review, we provide an overview of the AR side of the reality-virtuality continuum, and try to complement their findings for a comprehensive overview of AR, VR, and MR technology in automated driving. In our opinion, this abundance of current surveys in all these adjacent areas underscores the importance and timeliness of the topic.

1.4 Novelty and Contribution

To identify the latest trends in AR in AD research, we conducted a thorough, systematic literature review of 12 years of relevant publications between 2009 and 2020. We classified these papers based on their application areas, their utilized methodologies, and the level of vehicle automation used. In addition, we aim to provide impetus for standardization of methodology related to self-report, behavioral, and physiological measures in driving studies using AR technology.

The main contribution of this work is four-fold, regarding the status quo and derivation of future research directions:

- Identify the primary application areas for automotive user interface research using AR technology.
- Provide an overview of the methodologies and AR environments commonly utilized in driving automation research.
- Identify open challenges and opportunities, and propose recommendations for the research of AR in AD.
- Provide a set of recommendations and emerging possibilities for AR study design in driving automation research.

The rest of the paper is organized as follows: Section 2 details the method and process we followed to find, rate, and review the relevant papers. Section 3 then gives a high-level overview of the core relevant papers and studies, and establishes the classifications, followed by detailed reports on each of the classifications. Section 4 provides details of each of the classifications, highlighting some of the most impactful user studies as well as challenges for each application area. Finally, Section 5 concludes by summarizing the review and providing an outlook for future research.

2 Methodology

Based on the current state of research, the topics of interest for augmented reality HCI research in automated driving, and the authors' experience in AR and AD research, we determined the review process for this literature review. For the analysis of related literature, we examined publications at the intersection of augmented and virtual reality, human-computer interaction, and automated driving.

We followed a three-step process, which are **search** (identify relevant candidate publications), **rate** (determine the relevance of all candidates), and **code** (extract important characteristics of the core relevant papers) in our literature review. Although these steps were mainly performed sequentially, we made several repetitions and overlaps to ensure their validity.

Search: For exploring the body of work in AR in AD, we conducted a thorough literature research using the standard scientific search engines and databases such as the ACM Digital Library, IEEE Xplore, Science Direct, Wiley Online Library, and Google Scholar. Additionally, besides being a comprehensive source of AR and AD research, they contain impactful conferences and highly ranked journals such as CHI, DIS and TOCHI, VR, VRST, ISMAR, AutomotiveUI, and IV. We based our search on combinations of the keywords “augmented

reality,” “virtual reality,” “mixed reality,” “driving,” “automated vehicle,” and “vehicle automation.” After an initial round of selections, we found that some papers that use (screen-fixed) augmented reality using HUDs or WSDs did not include the keywords “augmented reality.” We therefore expanded our search parameters with the keywords “head-up displays/HUDs” and “windshield displays/WSDs”:

“augmented reality” OR “virtual reality” OR
 “mixed reality” OR “HUD/WSD”
 AND
 “driving” OR “automated vehicle” OR
 “vehicle automation”

In addition, we focused our search on papers that integrate aspects of AR in AD and have been presented in AR/VR visualization, human-computer interaction, and automotive user interfaces conferences and journals. The basis for the selection process included original research papers published in the respective journals between 01/01/2009 and 31/12/2020 (inclusive). Furthermore, we investigated recent issues of journals and conferences in these fields to search for papers with a closely related topic that may have escaped our previous search. The literature corpus also included literature from an existing list based on authors’ previous work. The decision on inclusion in the candidate list was based primarily on the keywords, title, and abstract to initially reduce the number of papers to a reasonable number while ensuring that no potentially relevant work was excluded. We also excluded the current year (2021) because the publication year had not yet been completed and thus not all potentially relevant papers were available. In addition to the papers found in this initial search, some papers were later found to be relevant, for example, by analyzing the related papers of the reviewed publications. Finally, this step resulted in a total number of 310 candidate papers that were considered interesting for further analysis.

Rate: For an initial classification of the retrieved literature, the entire set of 310 candidate papers was rated for their relevance to AR in AD. Prior to the rating process, the authors discussed and agreed upon initial relevance

criteria. Following the establishment of these initial criteria, each paper was analyzed to determine whether it was of interest for further consideration in the list of core relevant papers. These ongoing meetings during the rating phase also helped to further refine and clarify the relevance criteria and to standardize the way they were used by the authors. For the entire body of literature, the criteria for inclusion depended on the quality and provision of details about the user study. For example, publications that provided only an abstract or no detailed description were excluded. Other criteria for relevance were based on the type of work, so, for example, papers that addressed pedestrian safety using augmented reality technologies were considered relevant based on whether the system was used in the context of automated driving. Papers addressing interaction techniques or devices, as well as user studies and other evaluations or benchmarks, were considered in terms of the quality or usefulness of the particular interaction modality or study in the context of AR in AD. In addition, literature reviews and legal, theoretical, or ethical papers (e.g., Inners & Kun, 2017) were not included. Finally, from the set of eligible papers, we identified 199 core relevant papers, with 156 papers reporting user studies selected for further analysis.

Code: After the selection of papers, we devised a reviewing strategy for the literature analysis in order to develop a standardized reviewing approach. Using the PRISMA flow diagram (Moher, Liberati, Tetzlaff, Altman, et al., 2010), shown in Figure 2, we decided on the inclusion or exclusion of each paper in a standardized and stepwise manner. Moreover, we held several brainstorming sessions to categorize the subtopics to be studied. Based on the keywords and abstract of each paper as well as the current and historical paper tracks of the AutomotiveUI conference, the “premier forum for UI research in the automotive domain” (AutomotiveUI, 2009), we categorized our body of literature into eight main application areas. These subtopics are reflected by the subsections in Section 4. In case a paper covers multiple application domains (e.g., navigation and trust), we classified according to the main intent of the paper. For example, if the effects of different navigation cues on driver trust and acceptance were studied, we would

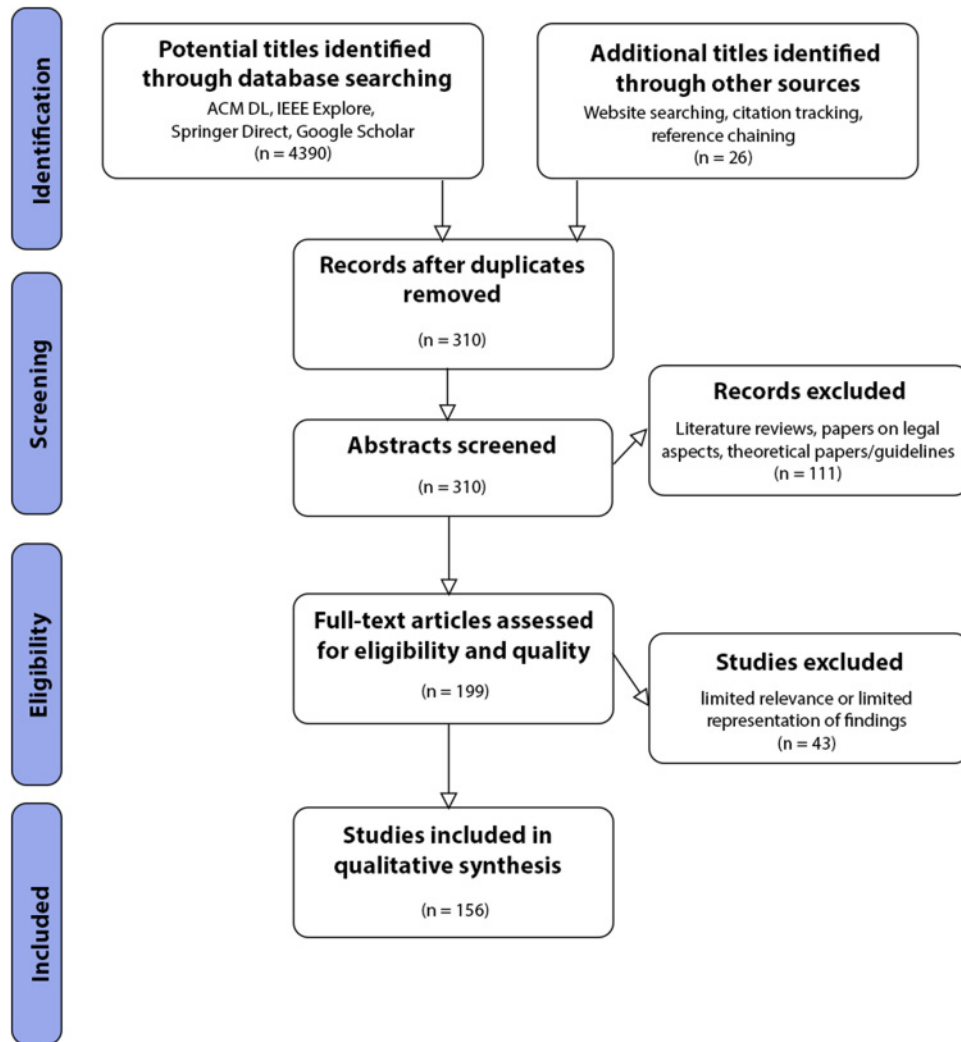


Figure 2. PRISMA flow diagram depicting the paper selection process.

classify that work (e.g., Wintersberger, Sawitzky, Frison, and Riener, 2017) in the application domain *Trust & Acceptance* rather than *Navigation & Routing*.

Subsequent to the categorization, we coded the core relevant papers using keywords extracted from the analyzed papers, or based on already existing taxonomies on the respective topic.

2.1 Limitations and Validity Concerns

Although we strived for methodology and rigor in selecting and reviewing these 156 papers, we acknowl-

edge several limitations and validity issues with our approach. First, we may have overlooked bibliographic sources. While using the bibliographic databases mentioned above has the advantage of covering a wide range of publication venues and topics, it is possible that we omitted venues and papers that we should have added. The second issue concerns the search parameters we used. Although the search terms we used appear to be genuine and carefully chosen, there may be papers that did not include the suggested keywords when describing automotive AR applications and usages. For example, articles may describe safety research evaluated with a

Table 1. *Summary of 156 Reviewed Papers*

Application Area	Paper	Mean	Publication		Device			Level of Automation		
		Author	Journal	Conference	HUD	HMD	Other	L0-2	L3-4	L5
Safety & Driver Assistance	58	3.88	17	41	20	5	34	48	7	2
Navigation & Routing	24	4.00	5	19	10	2	12	20	4	0
User Interface Design	33	4.27	7	26	12	6	16	19	7	2
Interaction Modalities	16	4.06	5	11	3	3	10	8	6	0
Passenger Experiences	5	4.60	1	4	1	2	3	1	2	1
eHMI & Vulnerable Road Users	6	4.00	1	5	2	4	0	1	0	3
Trust & Acceptance	9	4.67	3	6	1	3	5	1	0	8
Simulator & Motion Sickness	5	3.40	2	3	3	2	0	3	1	0
Overall	156	4.11	41	115	52	27	80	101	27	16

driving simulator that simulates augmented reality cues, but the term “augmented reality” is not included in the keywords, title, or abstract. Finally, we did not calculate the average citation count (ACC) for each paper. While this filter has been applied in previous literature reviews (e.g., Dey et al., 2018), we decided against excluding papers based on their impact, as some papers with seemingly narrow use cases and perhaps no immediately generalizable scope for AR in AD research may soon prove instrumental.

3 High-Level Overview of Reviewed Papers

In total, we reviewed the user studies reported in 156 papers. Table 1 provides a summary statistics for these papers, and Table 2 gives an overview of the user study details.

3.1 User Studies over Time

Figure 3A shows the total number of AR papers published between 2009 and 2020 with in the automotive domain. The figure shows that the number of AR papers published in 2020 is a large increase over those published in 2009. Overall, the number of papers uti-

lizing augmented reality technology is growing at a fast pace.

3.2 Study Design

As displayed in Table 2, most of the papers (115, or 74%) used a within-subjects design, 25 papers (16%) applied a between-subjects design, and 15 papers (10%) employed a mixed-subjects design. For 10 studies, it was not reported or clear from the study details which study design they reported.

3.3 Study Type

We found that it was rare for researchers to report details on pilot studies before their main study. Only 12 papers (7%) reported running a pilot study in their experimentation process. Only 6 papers provide just the results of the pilot study, without further main studies, which indicates that the importance of pilot studies is not well acknowledged. Most studies (131, or 80%) were performed in controlled laboratory environments, while 20 studies (13%) were conducted in a natural environment or as a field study. As most of the user studies were formal user studies with participants, and utilized no heuristic evaluation, this might point to heuristics and standardizations (Dey et al., 2018) needed to be

Table 2. Summary of the User Studies Reported by the 156 Reviewed Papers

Application Area	Study Type			Study Design			Study Location			Data Collected			Median Participants
	Lab	Field	Pilot	Within	Between	Mixed	Indoor	Outdoor	Both	Quant.	Qual.	Both	
Safety & Driver Assistance	48	7	3	36	15	7	50	7	1	22	5	31	28
Navigation & Routing	17	5	1	19	2	3	18	6	0	1	5	18	24
User Interface Design	29	4	3	25	4	4	27	4	2	9	8	16	24
Interaction Modalities	16	0	1	13	1	1	16	0	0	1	2	12	24
Passenger Experiences	4	1	2	4	1	0	4	1	0	0	3	2	24
cHMI & Vulnerable Road Users	6	0	1	5	1	0	4	2	0	1	3	2	12
Trust & Acceptance	9	0	1	8	1	0	9	0	0	0	6	3	26
Simulator & Motion Sickness	2	3	0	5	0	0	2	3	0	2	0	3	18
Overall	131	20	12	115	25	15	130	23	3	36	32	87	23

developed for future AR applications and use cases in the automotive domain. Due to design limitations, many AR studies use a simplified version of traffic scenes as well as their visual fidelity with respect to pedestrians and other road users, among others. While most works provide screenshots of the study environment (e.g., using VR driving simulators), it is often unclear to what extent the VR scene resembles the real world and real situations. Consequently, this circumstance negatively affects the external validity of user study results, as there is no guarantee that they can be transferred to real-world environments, as VR driving simulators may not be realistic enough due to limited field of view or lack of motion feedback. As Yeo, Kim, and Kim (2020) found, mixed reality configurations in real vehicles with motion feedback achieve the highest level of presence among simulators for automated driving. However, conditions in realistic driving tests are not as controlled and reproducible as in a laboratory environment and may differ in terms of surrounding traffic, weather conditions, or vehicle speeds. Therefore, with increasing computer and graphics power, we see the need to create “digital twins” of the real world (e.g., a highway segment, a city district) that provide a content-rich representation of the real world. The results of our review also show that most studies have conducted single interaction sessions and

therefore show only snapshots of initial use. Although usability has been shown to provide reliable information about user engagement within a single session (Forster et al., 2019), user experience, and trust may require a longer period of time to reach robust behavioral and attitudinal levels (Lee & Moray, 1992). Longitudinal studies should therefore consider these issues.

3.4 Data Type

We investigated the use of qualitative (subjective) and quantitative (objective) data types, and found that a total of 87 papers (56%) collected both quantitative and qualitative data, 36 (23%) papers only quantitative, and 32 (21%) only qualitative. The predominantly used quantitative data included driving performance (e.g., reaction time, braking distance), gaze and glance behavior, and physiological measurements, such as the heart rate variability and breathing metrics. Further, task performance (error/accuracy) was measured. As for subjective ratings, workload, user experience, simulator sickness, as well as custom self-ratings were reported. The specific types of data to be collected often depend on the application area. For example, quantitative measurements such as braking and take-over performance are required for safety applications, while self-assessed

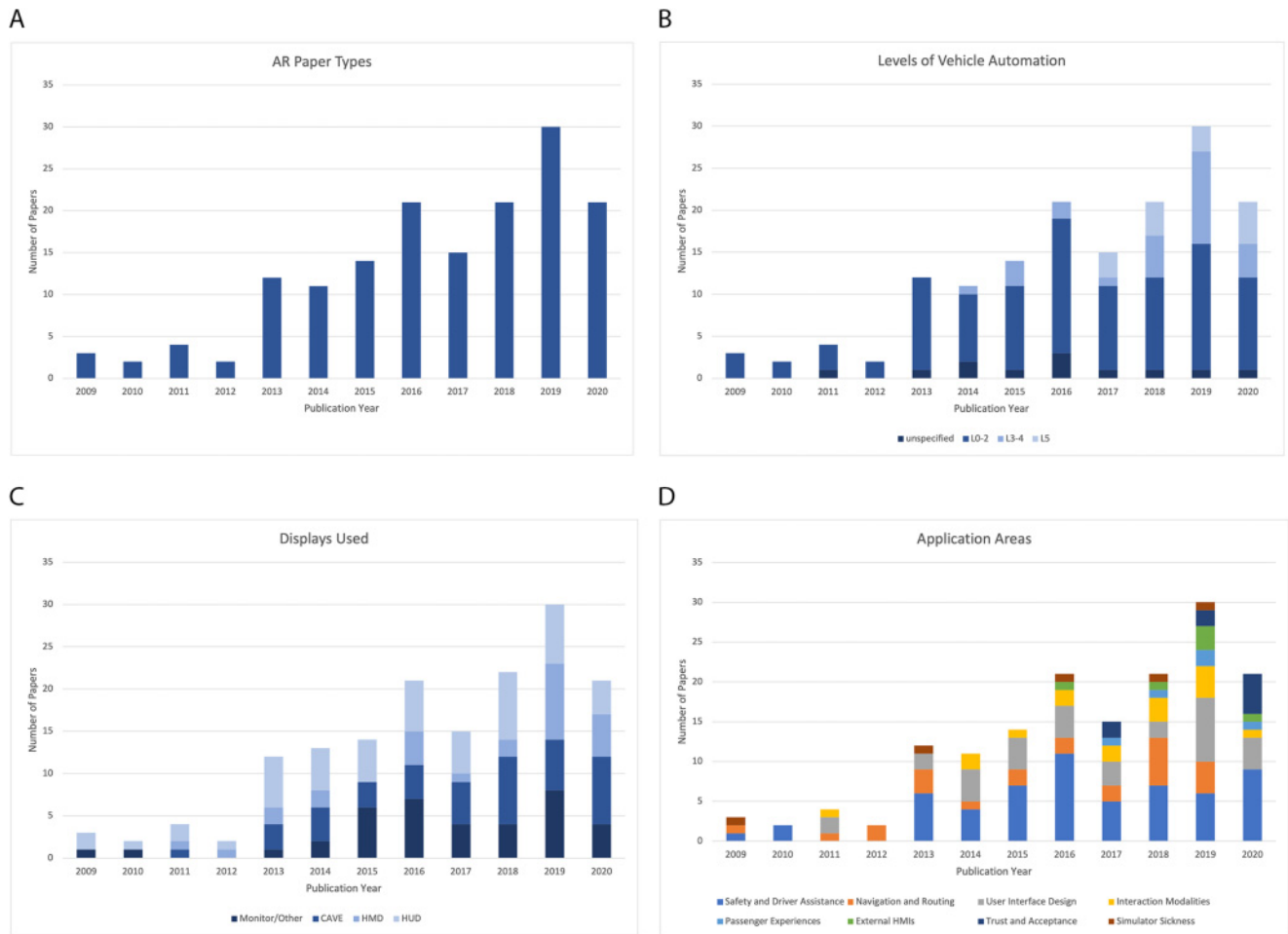


Figure 3. High-level overview of the analyzed papers. (A) AR papers utilized in automotive research. (B) Levels of vehicle automation. (C) Displays/devices used. (D) Categorization of the papers into eight application areas.

qualitative data such as acceptance and user experience data are more relevant for evaluating driver/passenger trust and well-being. In VR setups that simulate AR displays, at least one subjective measurement needs to be included, namely simulator sickness, which establishes the internal validity of the user study. In addition, quantitative data such as eye or head tracking data can nowadays be more easily collected with HMDs (such as the HTC Vive Pro Eye). In general, we recommend using both quantitative and qualitative data types for conditionally automated driving studies (SAE Levels 3 and 4), as they should also include take-over requests (i.e., measuring take-over performance). In addition,

task performance and situational awareness can be evaluated quantitatively, that is, effectiveness and efficiency according to ISO-9241 (ISO, 2018). Considering the user experience aspects, satisfaction should be included as an attitudinal component (ISO, 2018) using qualitative measurements such as self-rating questionnaires and interviews. For fully automated driving configurations (SAE level 5), the focus of user studies will switch to in-vehicle experiences such as transforming the vehicle into a mobile office or entertainment platform (Schartmüller et al., 2018), and therefore subjective ratings might be of greater importance. In general, for user studies, we recommend method triangulation as described by

Pettersson, Lachner, Frison, Riener, and Butz (2018), including questionnaires (both standardized and custom), interviews, and activity logging. While the research by Pettersson et al. (2018) focuses on user experience, we recommend the use of method triangulation for other application areas as well (see Section 4).

Further, the majority of the user studies were carried out in an indoor environment (130, or 83%), while outdoor studies were run in 23 cases (15%), and 3 studies reported a combination of both settings (2%). These findings seem plausible since simulated/virtual environments are used for the benefit of study participants' safety. Beyond that, the use of augmented reality technology in road traffic scenarios is currently still limited by the confined field of view of HMDs/HUDs, and the detection and registration of spatial objects.

3.5 Participants

Analysis of the demographic data of the study participants identified that most of the studies were conducted with young participants from university populations. The average age of participants in the user studies was 30 years. However, while most user studies included male and female participants (78%, some studies did not explicitly report gender), the ratio of female to male participants was quite low (34% of total participants where a distinction between male and female subjects was reported). Regarding the source of recruitment, we found that most participants were recruited from universities (26%), whereas recruitments from the public (10%) and private sector (5%) were relatively uncommon. In the remaining studies, the source of recruitment was not explicitly stated, suggesting that many researchers may not know how to report the details of user studies, limiting their reproducibility and validity. Participant analysis indicates that many automotive AR user studies are conducted with young male university students, as contrasted with a more diverse and representative sample of the population. In our literature review, we only distinguish between male and female participants, as we did not find any papers in our literature corpus that reported other genders. However, we encourage researchers and practitioners to include and report other

genders to promote inclusion. Because most of the reports were of young, male students, the study results are inherently biased, limiting the external validity of the results. In addition, the technical and technological affinity of these participants is typically much higher than that of the general public, which may have a significant impact on their performance. We therefore urge experimenters to make more efforts to recruit female and other participants, as well as different age groups, cultural backgrounds, and levels of technology affinity, to reduce the bias of the results and promote their generalizability (Howard & Borenstein, 2018).

3.6 AR Devices

Figure 3C gives an overview of devices used in the reported user studies, categorized into immersive HMDs, HUDs/WSDs, and CAVEs, as well as non- to semi-immersive monitor/smartphone setups. While the use of immersive AR/VR displays and study setups has continuously increased since 2013, the use of less immersive displays has declined. This shows that the availability of advanced MR devices for researchers has improved. Overall, 52 papers (33%) explored HUDs/WSDs, 27 papers (17%) reported using HMDs, 42 (27%) utilized CAVEs, and less immersive setups were used by 38 (24%) of papers. This results in more than 100% as several studies utilized multiple/combined AR/VR setups. A more detailed overview of utilized devices can be found in each application area in Section 4. While HMDs have a cost and portability advantage, CAVEs are not constrained by a limited field of view because projections are used to create the VR environment. Over the last decade, AR technology has become more affordable and easier to set up. As a consequence, the use of non-immersive setups, including monitors, should be discouraged, especially for continuous-depth applications such as simulated AR head-up or windshield displays.

3.7 Levels of Vehicle Automation

Figure 3B visualizes the total number of papers we examined for the years 2009 to 2020, categorized by

the researched level of vehicle automation, from manual (SAE L0) to fully automated driving (SAE L5). AR research on conditionally automated driving (SAE L3–4) was reported starting in 2015, when AR/VR hardware of suitable quality for research emerged. No vehicle automation was reported by 102 papers (65%). Conditionally automated driving (SAE level 3 and 4) was researched by 27 papers (17%), fully automated driving (SAE level 5) by 10%, and research where the level of vehicle automation was not critical, and therefore not reported, was covered by 8% of the reviewed papers. Seemingly, the use of AR technology (Figure 3A) and research into higher levels of vehicle automation appear to move forward hand-in-hand, which is consistent with Gartner's hype cycle for emerging technologies (Gartner, Inc., 2014).

3.8 Categorization

We organized the 156 papers into a total of eight application areas, which shows that there is a wide and open field for automotive researchers using AR technology. Figure 3D provides an overview of how these application areas have evolved over time. Initial research with AR in the automotive domain was primarily related to safety/driving assistance and navigation aspects. More recently, research has focused on the experiences of passengers, vulnerable road users such as pedestrians and cyclists, and the trust and acceptance of automated vehicles. In addition, interaction concepts inside and outside the vehicle (external human-machine interfaces, eHMIs) are being investigated.

3.9 Further Studies Information

As displayed in Table 1, most application areas had a similar mean number of authors per paper, ranging between 3.40 (Simulator and Motion Sickness) and 4.00 (Navigation and Routing, eHMI and Vulnerable Road Users). However, papers in the Trust and Acceptance area had the highest average number of authors (4.67), illustrating the interdisciplinary research field (automotive, human-computer interaction, AR/VR visualization, and psychology). Most of the papers were pub-

lished in conferences, while for some application areas (Trust and Acceptance, Simulator and Motion Sickness) many studies were published in journals (approx. 50%).

In total, the 156 papers reported 170 studies (Table 1), most of which (145, or 93%) reported only one user study. The remaining papers reported two or more studies, including pilot studies. In terms of median number of participants, we found that the application areas Safety and Driver Assistance (28), and Trust and Acceptance (26) had the highest number of participants per study. Other application areas used approx. 24 participants per user study. Only studies on external HMIs were conducted with fewer participants at the median (12). Therefore, the use of 24 to 28 participants in user studies seems to be a typical range in the AR automotive community.

In the following section, we review user studies in each of the eight application areas separately. We additionally present overview tables, and discuss each application area by examining already conducted research, so that readers can gain more detailed insights into user studies from that domain, and find research gaps as well as potential scopes for future research.

4 Augmented Reality Application Areas

4.1 Safety and Driver Assistance

In the Safety and Driver Assistance application area, 58 studies were reported. The majority of these studies explored safety aspects for manual driving (see Table 3), although seven studies examined highly automated driving, and one study evaluated mixed traffic situations, where both manual and automated vehicles share the road. Interestingly, only two studies (Lindemann, Lee, & Rigoll, 2018b, 2018c) investigated safety in the context of fully automated vehicles. Most studies regarded situation awareness and crash warnings, however, also smartphone usage and take-over requests were evaluated. Out of the 58 studies, 19 reported using CAVEs, 20 used HUDs, five used HMDs, and 15 used some form of monitor/smartphone display. While 36 studies were carried out as within-subject studies, 15

Table 3. Summary of User Studies in Safety and Driver Assistance Application Area

References	Topic	Displays used	Vehicle		Dependent measures	Participants (female)
			Automation	Data type		
Ng-Thow-Hing et al. (2013)	AR driving aid	HUD	L0-2	O+S	Error/Accuracy, Interview	16 (8)
Lauber and Butz (2013)	HMD view management	HMD	L0-2	O+S	Reaction Time, Error/Accuracy, Rating, Interview	35 (NA)
Rusch et al. (2014)	AR cues for elderly drivers	CAVE	L0-2	O	TTC, Time, Error/Accuracy	64 (29)
Lauber and Butz (2014)	Head-stabilized warnings	HUD, HMD	L0-2	O+S	Reaction Time, Rating	25 (NA)
Eyraud et al. (2015)	Allocation of visual attention	Monitor	L0-2	O	Gaze	48 (21)
Morley et al. (2016)	Smartphone use	HMD	L0-2	S	Rating, Interview	22 (14)
Politis et al. (2015)	Multimodal displays for take-overs	Monitor	L3-4	O+S	TOR Time, Time, Distance, Error/Accuracy, Rating	21 (3)
Smith, Doutecheva et al. (2015)	HUD depth judgement of real world objects	HUD	L0-5	O	Error/Accuracy	24 (NA)
Wulf et al. (2015)	Situation awareness	CAVE	L3-4	O	Reaction Time, Collisions, Gaze	135 (NA)
Langlois and Soualmi (2016)	AR take over	CAVE	L3-4	O	TOR Time, Distance, Acceleration/Braking	26 (0)
Kane et al. (2016)	AR side view mirrors	Monitor	L0-2	O	Error/Accuracy	16 (NA)
Hwang and Kim (2016)	AR HUD psychological characteristics	HUD	L0-2	O+S	Time, Driving Behavior Determinants Questionnaire	31 (0)
Phan et al. (2016)	Driver awareness	Monitor	L0-2	O+S	Time, Distance, Acceleration/Braking, Rating	27 (4)
Wang and Soffker (2016)	Driving efficiency interface	Monitor	L0-2	O+S	Speed, Gaze, NASA TLX	32 (3)
Sánchez et al. (2016)	HUD for improving driving efficiency	HUD	L0-2	S	Rating	20 (NA)
Lorenz et al. (2014)	AR take-over scenarios	CAVE	L3-4	O	TOR Time, Acceleration/Braking, Steering, Gaze	46 (10)
Chen et al. (2017)	Eliminating driving distractions	HUD	L0-2	S	Rating	15 (NA)
Lindemann and Rigoll (2017b)	See-through cockpit	CAVE	L0-2	O	Distance	8 (3)
Lindemann and Rigoll (2017a)	Diminished reality	CAVE	L0-2	O+S	Steering, View Angle, Distance, Rating	8 (3)
Lindemann et al. (2018b)	Situation awareness	CAVE	L5	O+S	Error/Accuracy, Rating	32 (8)
Park et al. (2018)	Driving assistance information in connected vehicle environments	HUD	L0-2	S	Rating	48 (24)

Table 3. *Continued.*

References	Topic	Displays used	Vehicle		Dependent measures	Participants (female)
			Automation	Data type		
Lindemann et al. (2018c)	Situation awareness	HUD	L5	O+S	Error/Accuracy, Rating	32 (8)
Frémont et al. (2019)	Driver awareness	Monitor	L0-2	O+S	Time, Acceleration/Braking, Rating	23 (4)
Utsumi et al. (2019)	On-road virtual visual references	HUD	L0-2	O	Distance, Gaze	34 (27)
Lindemann et al. (2019)	Short notice take-overs AR warning	CAVE	L3-4	O+S	Reaction Time, Distance, Steering, NASA TLX, Rating	18 (8)
Amersfoorth et al. (2019)	Translucency awareness cues	Monitor	L0-2	O+S	Reaction Time, Rating	20 (10)
Pichen et al. (2020)	Driver awareness on cyclists	CAVE	L0-2	O	Distance	30 (11)
Kraft, Maag, Cruz et al. (2020)	Maneuver coordination	CAVE	L0-2	O+S	Acceleration/Braking, Collisions, Gaze, Rating	68 (32)
Karatas et al. (2020)	Driver awareness	Monitor	L0-2	O+S	Speed, Gaze, Rating	36 (10)
Merenda et al. (2017)	Driver awareness	CAVE	L0-2	O+S	Time, Distance, Acceleration/Braking, Gaze, Rating	24 (9)
Bozkir et al. (2019)	Driver attention	HMD	L0-2	O	Distance, Acceleration/Braking, Gaze	16 (4)
Calvi et al. (2020a)	Driver awareness	CAVE	L0-2	O+S	TTC, Distance, Acceleration/Braking, Rating	46 (13)
Persson et al. (2020)	Mitigate blind spots	HMD	L0-2	O+S	Error/Acceptance, NASA TLX, Rating	15 (3)
Zimmermann et al. (2018)	Cooperative driving through social interaction	Monitor	L0-2	O+S	Time, Rating	39 (8)
Doshi et al. (2009)	Driver assistance	HUD	L0-2	O	Time, Gaze	11 (NA)
Langner et al. (2016)	Traffic awareness driver assistance	HUD	L3-4	O	Gaze	NA (NA)
Fröhlich et al. (2010)	Safety information	Smartphone	L0-2	O+S	Error/Accuracy, Rating, Interview	28 (12)
Saffarian et al. (2010)	Cooperative adaptive cruise control	HUD	L0-2	O+S	Time, Acceleration/Braking, Rating Scale for Mental Effort	20 (10)
Rusch et al. (2013)	Driver attention using AR cues	HUD	L0-2	O	Reaction Time, Error/Accuracy	27 (14)
Fu et al. (2013)	AR warnings	CAVE	L0-2	O	Collisions, TTC, Reaction Time, Acceleration/Braking	24 (10)
Kim et al. (2013)	Crash warning system	CAVE	L0-2	O+S	Collisions, Reaction Time, NASA TLX, Rating	16 (8)
Alves et al. (2013)	Forward collision warning	Monitor	L0-2	O+S	Reaction Time, Distance, Rating	22 (4)

Table 3. *Continued.*

References	Topic	Displays used	Vehicle		Dependent measures	Participants (female)
			Automation	Data type		
Kazazi et al. (2015)	Visual warning HUD	HUD	L0-2	O	Collisions, Reaction Time, Acceleration/Braking	72 (23)
Häuslschmid, Schnurr et al. (2015)	AR WSD warning	Monitor	L0-2	O+S	Distance, Time, Error/Accuracy, Gaze, Rating	18 (2)
Winkler et al. (2015)	Driver gaze and driving behavior	HUD	L0-2	O	Collisions, Reaction Time, Acceleration/Braking, Gaze	60 (27)
Kim et al. (2016)	AR pedestrian warning	HUD	L0-2	O	Reaction Time, Distance, Acceleration/Braking	16 (NA)
Schwarz and Fastenmeier (2017)	AR warnings	CAVE	L0-2	O+S	Acceleration/Braking, Gaze, AttrakDiff, Rating	88 (18)
Hwang et al. (2017)	Reduced visibility traffic environment	HUD	L0-2	O+S	Reaction Time, Reckless Driving Behavior Questionnaire	31 (0)
Kim et al. (2018)	Pedestrian collision warning	HUD	L0-2	O	Gaze, Distance, Acceleration/Braking	16 (NA)
Wu et al. (2018)	Crash warning in fog conditions	Monitor	L0-2	O	TTC, Response Time, Acceleration/Braking	54 (NA)
Winkler et al. (2018)	Warning types	CAVE	L0-2	O+S	Collisions, Acceleration/Braking, Rating	60 (27)
Utsumi et al. (2018)	Emergency behavior	CAVE	L3-4	O+S	Error/Accuracy, Rating	39 (27)
Smith et al. (2020)	Driver response to unexpected events	HUD	L0-2	O+S	Collisions, Gaze, Rating	20 (10)
Calvi et al. (2020b)	AR for left turn maneuvers	CAVE	L0-2	O	Error/Accuracy, Time	46 (13)
Li, Chen et al. (2020)	Driving safety experience comparison	HUD	L0-2	O+S	Time, Gaze, Heart Rate, NASA TLX, UEQ	60 (NA)
Bram-Larbi et al. (2020)	Collision avoidance HUD	CAVE	L0-2	S	Rating	50 (10)
Hwang et al. (2016)	Risk perception	Monitor	L0-2	O+S	Response time, Rating	31 (1)
Charissis (2014)	AR HUD warning	Monitor	L0-2	O+S	NA	NA (NA)

studies were designed as between, and seven as mixed (within and between design). On average, 28 participants joined a within-subject study, with approximately one third of the subjects being female. Most studies were carried out in indoor locations (50), seven were carried out outside, and one study was run in both location settings. About half of all studies reported both objective and subjective data, while the rest focussed on

mostly quantitative results. Regarding objective data, error/accuracy, collisions, distance, time, such as take-over request time (TOR), and time to collision (TTC), as well as gaze metrics were reported prominently, while subjective feedback included user experience (User Experience Questionnaire, UEQ) and workload (NASA Task Load Index, NASA TLX), among custom questionnaires.

4.1.1 Discussion. The Safety and Driver Assistance domain is dominated by research on manual driving, encompassing the visualizations of warnings related to the road environment, and collision avoidance (Karatas et al., 2020; Schwarz & Fastenmeier, 2017; Wu, Abdel-Aty, Park, & Zhu 2018). However, as mixed traffic situations will become more common in the future, we believe that research interest will increase for highly automated vehicles, for example, when it comes to cooperative driving, and teleoperated driving (Zimmermann et al., 2018). VR also plays an important role in safety research, as AR awareness/attention cues can be simulated. As such, new opportunities will emerge encompassing both AR and VR in automotive safety research, including gamification concepts (Schroeter & Steinberger, 2016). We encourage researchers to include vulnerable road users in AR safety research, including pedestrians (Kim et al., 2016) and cyclists (Pichen, Yan, & Baumann, 2020). Additionally, AR automotive research should include take-over performance and safety aspects (Morales-Alvarez, Sipele, Léberon, Tadjine, & Olaverri-Monreal, 2020).

Particularly, the transition from NDRTs visualized on AR displays to manual driving should further be explored. To this end, keeping the driver in the loop must be a precondition for AR interfaces in conditionally automated driving. We further encourage researchers to investigate world-fixed AR using (simulated) windshield displays (Saffarian, Happee, Abbink, & Mulder, 2010) and their potential to increase situation awareness without cognitively overloading the driver/passenger.

4.2 Navigation and Routing

A total of 24 studies were reported in the Vehicle Navigation and Routing application area. The majority of the studies investigated navigational aids for manual driving (see Table 4), although Schömig et al. (2018) presented an AR interface for partially automated vehicles. Interestingly, out of the 24 studies, eight reported using CAVEs, four used a monitor/smartphone setup, two used HMDs, and ten used a HUD display. This seems intuitive as navigational aids usually necessitate a full view of the surrounding (virtual) environment,

which CAVEs and HMDs provide. For smaller navigation cues, the limited field of view was adequate. While 19 studies were designed to be within-subjects, two studies were designed as between, and three as mixed. On average, 24 participants joined a within-subject study, with approx. 28% of the subjects being female. Most studies were carried out in indoor locations (18, or 75%), and the rest was conducted outdoors. 75% of studies reported both objective (quantitative) and subjective (qualitative) data, while the rest focussed on qualitative results. Regarding objective data, error/accuracy and gaze metrics were reported prominently, while subjective feedback included interviews, self-ratings, NASA TLX, and perceived safety.

4.2.1 Discussion. Research conducted in this application area is mostly related to manual (Moussa, Radwan, & Hussain, 2012; Topliss, Pampel, Burnett, Skrypchuk, & Hare, 2018) and conditionally automated vehicles (Feierle, Beller, & Bengler, 2019; Kunze, Summerskill, Marshall, & Filtneß, 2018). With the advances in HMD and CAVE technology, navigation, and routing tasks can be explored more immersively than with monitor-based setups. With respect to higher levels of vehicle automation, more research is needed on motion sickness (see Section 4.8), as navigation aids in combination with AR HUDs or WSDs could be beneficial. In addition, the emergence of AR technology can help transform conventional two-dimensional directional instructions into three-dimensional instructions that blend into the external environment, such as guided arrows (Tangmanee & Teeravarunyou, 2012) or road markings (Wiesner, Ruf, Sirim, & Klinker, 2017). A new future research direction for automated vehicles would be to integrate navigation tasks with passenger experiences and gamification (see Section 4.5). Schroeter and Steinberger (2016), for example, utilize AR to playfully alert drivers to situational warnings, a concept that could be applied to navigation and route planning.

4.3 User Interface Design

A total of 29 studies were reported in the User Interface (UI) and Human-Machine Interface (HMI)

Table 4. Summary of User Studies in Vehicle Navigation and Routing Application Area

References	Topic	Displays used	Vehicle		Dependent measures	Participants (female)
			Automation	Data type		
Kim and Dey (2009)	AR map on WSD, elderly drivers	Monitor	L0-2	O+S	Error/Accuracy, Rating, Gaze, Driving time	24 (13)
Medenica et al. (2011)	Comparing of navigation aids	CAVE	L0-2	O+S	Error/Accuracy, Rating, NASA TLX, Gaze	18 (NA)
Moussa et al. (2012)	Left-turn maneuver	HMD	L0-2	O+S	Time, Acceleration/Braking, Rating	44 (18)
Tangmanee and Teeravarunyou (2012)	HUD displaying guided arrows	HUD	L0-2	O+S	Time, Gaze, Rating	5 (2)
Tran et al. (2013)	Projected vehicle paths	HUD	L0-2	S	Rating	4 (1)
Kawamata et al. (2013)	Lifted road map view	HUD	L0-2	S	Rating	6 (NA)
Palinko et al. (2013)	AR navigation	CAVE	L0-2	O+S	Error/Accuracy, Rating, NASA TLX, Gaze	10 (NA)
Bark et al. (2014)	3D volumetric HUD navi	Monitor	L0-2	O+S	Time, Gaze, Error/Accuracy, Rating	9 (2)
Pfannmüller et al. (2015)	HUD navigation concepts	HUD	L0-2	S	Rating	30 (7)
Bolton et al. (2015)	Landmark-based navigation	CAVE	L0-2	O+S	Error/Accuracy, Rating, NASA TLX, Gaze	20 (6)
Jose et al. (2016)	Comparison AR HUD, HDD, HMD	HMD	L0-2	O+S	Error/Accuracy, Rating	18 (4)
Yin et al. (2016)	In-vehicle traffic lights	Monitor	L0-2	O	Time, Acceleration/Braking, Gaze	11 (2)
Wiesner et al. (2017)	Future road course display	HUD	L0-2	O+S	Acceleration/Braking, Gaze, Error/Accuracy, Rating	20 (7)
Yang et al. (2017)	In-vehicle traffic lights	CAVE	L0-2	O+S	Speed, Acceleration/Braking, Gaze, Rating	23 (0)
Cao et al. (2018)	AR navigation using a smartphone	Smartphone	L0-2	S	Rating	50 (10)
Schömig et al. (2018)	AR navigation	HUD	L3-4	O+S	Gaze, Rating	24 (NA)
Kunze et al. (2018)	Communicating uncertainty information	HUD	L3-4	O+S	Time, Rating	46 (16)
Merenda et al. (2018)	Goal-directed driving task	HUD	L0-2	O+S	Time, Error/Accuracy, Gaze, Rating	24 (7)
Shahriar and Kun (2018)	Camera-view AR	CAVE	L0-2	O+S	Error/Accuracy, Rating, NASA TLX, Gaze	18 (0)
Topliss et al. (2018)	Virtual lead vehicle as navigational aid	CAVE	L0-2	O+S	Error/Accuracy, Rating, NASA TLX	34 (8)
Schneider et al. (2019)	AR HUD navigation concepts	HUD	L0-2	S	Interview	24 (5)
Feierle et al. (2019)	Evaluating AR HUD in urban driving	CAVE	L3-4	O+S	Gaze, NASA TLX, Rating	30 (10)
Faria et al. (2019)	Spatial knowledge acquisition	HUD	L3-4	O+S	Error/Accuracy, NASA TLX	24 (NA)
Kraft, Maag, and Baumann (2020)	Cooperative driving	CAVE	L0-2	O+S	Error/Accuracy, Rating, Gaze, Perceived Safety	24 (13)

Design application area. We found that HUD and WSD concepts were most prominently researched (see Table 5). Regarding vehicle automation, all manual, conditional, and fully automated driving scenarios were examined. Further, there was no clear preference towards employed devices, as HMDs, CAVEs, and monitors/HUDs were utilized. While 25 studies were designed to be within-subjects, four studies were designed as between, and four as mixed. On average, 24 participants joined a within-subject study, with 40% of the subjects being female. With the exception of four studies, all studies were carried out in indoor locations. The majority of all studies reported both objective and subjective data, while the rest focussed on quantitative and qualitative results, separately. Regarding objective data, reaction times, error/accuracy, and gaze metrics were reported prominently, while subjective feedback included workload using NASA TLX, user experience using AttrakDiff, Simulator Sickness Questionnaire (SSQ), and IGroup Presence Questionnaire (IPQ), as well as self-ratings.

4.3.1 Discussion. Another application area of automotive augmented reality is the rapid UI design and prototyping process that can be achieved while being immersive and cost-effective, as Gerber et al. (2019) and Riegler, Riener, and Holzmann (2019b) demonstrate. VR driving simulators can be utilized to prototype such AR designs (Charissis, Papanastasiou, Chan, & Peytchev, 2013; Gerber, Schroeter, Xiaomeng, & Elhenawy, 2020; Morra, Lamberti, Prattico, Rosa, & Montuschi, 2019; Riegler, Weigl, Riener, & Holzmann, 2020c). Merenda et al. (2019a, 2019b) underline the importance of the level of detail and visual fidelity the virtual environment, as driving performance and gaze behavior are impacted by these parameters. While initial research efforts on automotive UI/HMI design focussed on displaying AR warnings to the driver (see Section 4.1), more recent research emphasizes the increased level of vehicle automation, and their opportunities and challenges. For example, Riegler, Wintersberger, Riener, and Holzmann (2019c) investigated what kind of content potential drivers of conditionally/fully automated vehicles

would use, where they should be placed, and what transparency levels are suitable. Additionally, screen-fixed and world-fixed content presentations are being explored (Riegler, Riener, & Holzmann, 2019a). Especially for SAE L3 and L4 vehicles, the provision of take-over requests must be visualized properly (Li, Schroeter, Rakotonirainy, Kuo, & Lenné, 2020). We noted that both qualitative and quantitative evaluations have been conducted for this application area, and we continue to advocate this, given the need to consider the performance of specific HMIs, as well as the corresponding subjective ratings. In this respect, personalization should be further explored, as the advent of more and more digital interfaces in vehicles and the simultaneous phasing out of hardware user interfaces (e.g., buttons, sliders) will enable novel tailored UIs that can be customized by drivers and passengers (Häuslschmid, Ren, Alt, Butz, & Höllerer, 2019). Additionally, we encourage to conduct more research on AR HUD/WSD UI designs in real-world settings as VR simulators might not be able to fully simulate AR content. Pettersson et al. (2019) compared a VR setup to a field study, and found that while VR studies do not fully replace field studies, they can provide added value to driving simulators and desktop testing. In particular, VR can be used for conveying the overall principles, or “the story” behind a user study. Further topics of interest in this application area are the design of non-driving related tasks (e.g., virtual button sizes, indicators, etc.), content/app switching (i.e., switching from one WSD application to another), and handover strategies, such as switching from content displayed on a mobile phone to the windshield display, and vice versa.

4.4 Interaction Modalities

We reviewed a total of 16 studies in the Interaction Modalities application area. Our findings show that research was conducted for both lower and higher levels of vehicle automation (see Table 6), and three studies reported details for all levels of vehicle automation. All AR/VR devices (HUDs, HMDs, CAVEs, monitors) were employed. Additionally, all but three studies collected both qualitative and quantitative data.

Table 5. *Summary of User Studies in User Interface Design Application Area*

References	Topic	Displays used	Vehicle		Dependent measures	Participants (female)
			Automation	Data type		
Yeh et al. (2013)	AR HUD visual perception	HUD	L0-5	S	Rating	300 (100)
Weinberg et al. (2011a)	HUD list selection	HUD	L0-2	O+S	Time, Distance, Steering, Gaze, NASA TLX, Rating	24 (10)
Weinberg et al. (2011b)	HUD list selection	HUD	L0-2	O+S	Time, NASA TLX, Rating	24 (10)
Fujimura et al. (2013)	3D HUD visual feedback	HUD	L0-2	O	Error/Accuracy	16 (8)
Jakus et al. (2014)	HUD multimodal display	CAVE	L0-2	O+S	Time, Collisions, NASA TLX, UEQ, Rating, Interview	30 (9)
Rao et al. (2014)	In-vehicle AR	CAVE	L0-5	S	Interview	214 (NA)
Soro et al. (2014)	Augmented video	CAVE	L0-2	S	Interview	11 (NA)
Lauber et al. (2014a)	HUD infotainment shortcuts	HUD, HMD	L0-2	O+S	Time, Distance, Error/Accuracy, NASA TLX, Rating	37 (9)
Kim et al. (2015)	AV user experience concepts	Monitor	L0-2	S	Interview	24 (8)
Smith, Streeter et al. (2015)	Visual search tasks	HUD	L0-2	O+S	Time, Distance, Error/Accuracy, Gaze, NASA TLX, Rating	16 (8)
Merenda et al. (2016)	HUD color naming and matching	HUD	L0-5	O	Time, Error/Accuracy	12 (6)
Häuslschmid et al. (2016)	WSD view management concept	HMD	L0-2	S	Interview	21 (6)
Smith et al. (2016)	HUD vs HDD comparison	CAVE	L0-2	O+S	Distance, Speed, Gaze, Error/Accuracy, NASA TLX, Rating	24 (NA)
Häuslschmid et al. (2017)	WSD text & shape recognition	Monitor	L0-2	O	Reaction Time, Time	46 (11)
Liu et al. (2017)	Attentional bias in WSD	HUD	L5	O	Error/Accuracy	18 (6)
Smith et al. (2017)	Eye scan patterns and HUD perception	HUD	L0-2	O+S	Time, TTC, Distance, Gaze, Error/Accuracy, NASA TLX, Rating	16 (8)
Lindemann et al. (2018a)	Explanatory WSD interface	HUD	L5	S	Rating	51 (21)
Bolder et al. (2018)	MR infotainment system	HMD	L0-2	O+S	Time, Error/Accuracy, SUS, IPQ, SSQ, Rating	30 (NA)
Riegler et al. (2019c)	WSD content types, size, location	Monitor	L3-5	S	Rating, Interview	63 (19)
Fereydooni et al. (2019)	AR and manual-visual task switching	HMD	L3-4	O	Time, Gaze, Error/Accuracy	5 (3)
Riegler et al. (2019a)	WSD content positioning	HMD	L3-4	O	TOR Time, Time, Error/Accuracy	23 (11)
Gerber et al. (2020)	NDRTs HUD vs Mobile	CAVE	L3-4	O	Gaze, Time	46 (23)

Table 5. *Continued.*

References	Topic	Displays used	Vehicle		Dependent measures	Participants (female)
			Automation	Data type		
Hensch et al. (2020)	Different display positions	CAVE	L3-4	O+S	Gaze, Error/Accuracy, Rating	50 (23)
Gabbard et al. (2020)	AR HUD color blending	HUD	L0-5	O	Time, Error/Accuracy	12 (6)
Li et al. (2020)	HUD vs HDD NDRTs take-over	CAVE	L3-4	O	Gaze	46 (23)
Merenda et al. (2019a)	Level of detail and visual fidelity	Monitor	L0-2	O+S	Time, Acceleration/Braking, Gaze, Rating	24 (NA)
Merenda et al. (2019b)	Level of detail and visual fidelity	Monitor	L0-2	O+S	Speed, Steering, Gaze, NASA TLX, Rating	12 (6)
Broy et al. (2014)	3D HUD design parameters	HUD	L0-5	O+S	Error/Accuracy, Rating	49 (12)
Häuslschmid et al. (2019)	Personalizing WSD content presentation	Monitor	L0-2	O+S	Reaction Time, Error/Accuracy, Rating	35 (13)
Häuslschmid, Osterwald et al. (2015)	WSD widgets & personalization	Monitor	L0-2	O+S	Error/Accuracy, Reaction Time, AttrakDiff, Rating	20 (6)
Topliss et al. (2020)	HUD time thresholds	CAVE	L0-2	O+S	TTC, Time, Gaze, Rating	60 (17)
Langlois et al. (2016)	Distance perception	CAVE	L0-2	S	Rating	24 (6)
Sportillo et al. (2019)	Autonomous driving training	HMD	L3-4	O+S	TOR Time, Rating	60 (31)

Quantitative criteria encompassed TOR times, error/accuracy, gaze, and steering metrics, among others.

Qualitative data included NASA TLX, SSQ, self-ratings, and interviews, among others. 13 studies were designed as within-subject studies, one as between-subject study, and one as mixed design. The within-subject studies were conducted with 24 participants on average, with one third of the subjects being female. With respect to the location, all studies were conducted indoors in controlled lab settings.

4.4.1 Discussion. Future intelligent vehicles will require suitable modalities for interacting with their UIs. Instead of more conventional hardware buttons, knobs, sliders, and touch screens, novel types of user interfaces (e.g., touch, speech, gestures, handwriting, and vision) (Norman, 2010) will be further explored in the automotive sector. VR driving simulators are commonly employed as they allow for rapid prototyping and iter-

ating through different interaction paradigms (Riegler, Riener, and Holzmann, 2019b; Schroeter & Gerber, 2018). There is growing research on driver-vehicle interaction that focuses on NDRTs rather than the primary driving task and uses an AR interface. Finger/hand gestures to interact with a HUD are being investigated (Brand, Büchele, & Meschtscherjakov, 2016; Lagoo, Charissis, & Harrison, 2019). In addition, Riegler, Aksoy, Riener, and Holzmann (2020a) explore head gaze-based interactions with WSDs to control an AR infotainment system. Eye gaze-based interactions with have also been investigated (Makedon, Konkol, Brandenburg, & Stark, 2020). Riegler, Weigl, et al. (2020c) make use of speech-based interactions to conduct a NDRT in conditionally automated vehicles. In addition, teleoperations need to be studied further in mixed traffic scenarios where shared fully automated vehicles might enter situations where human operators need to control the vehicle remotely (Bout, Brenden, Klingegård, Habibovic, & Böckle, 2017). This could be achieved

Table 6. *Summary of User Studies in Interaction Modalities Application Area*

References	Topic	Displays used	Vehicle		Dependent measures	Participants (female)
			Automation	Data type		
Wang et al. (2014)	Steering wheel speech interface	HUD	L0-2	O	Time, Distance, Error/Accuracy	24 (NA)
Lauber et al. (2014b)	HUD touch screen	HUD	L0-2	O+S	Time, Distance, Error/Accuracy, NASA TLX	70 (10)
Brand et al. (2016)	Pointing gestures at HUD	Monitor	L0-5	O+S	Pointing angle, Rating, Interview	10 (5)
Villalobos-Zúñiga et al. (2016)	Keyboard	Monitor	L0-2	O+S	Error/Accuracy, Time, Distance, Gaze, NASA TLX	16 (6)
Bout et al. (2017)	Teleoperation	HMD	L3-4	S	Rating	12 (2)
Lagoo et al. (2018)	Gestures	CAVE	L0-2	O+S	Error/Accuracy, Time, Rating	20 (7)
Weidner and Broll (2019)	Finger gestures	Monitor	L3-4	O+S	Agreement Rate, SSQ, Rating	23 (11)
Prabhakar et al. (2019)	Gaze and finger controlled HUD	HUD	L0-2	O+S	Distance, Time, Steering, Gaze, SUS, NASA TLX	49 (16)
Lagoo et al. (2019)	Gestures + HUD	CAVE	L0-2	O+S	Collisions, NASA TLX, Rating	20 (7)
Wang, Chen et al. (2019)	Gestures + HUD	Monitor	L0-5	S	Rating	20 (7)
Riegler et al. (2020a)	Head gaze	HMD	L3-4	O+S	NASA TLX, Time, Rating	24 (4)
Stockert et al. (2015)	Driver-automation interaction using feedback design	Monitor	L3-4	O+S	Reaction Time, Error/Accuracy, NASA TLX, Rating	21 (7)
Tippey et al. (2017)	Hands-free interaction with AR	Monitor	L0-2	O+S	Steering angle, Distance, Time, Error/Accuracy, Gaze, Rating	24 (9)
Schartmüller et al. (2018)	Keyboard Input	CAVE	L3-4	O+S	TOR Time, Time, Steering, TTC, NASA TLX, SSQ, Interview	20 (6)
Kun et al. (2019)	Conversations through AR	CAVE	L0-2	O+S	Distance, Gaze, Error/Accuracy, Rating	10 (2)
Qiu et al. (2011)	Accessibility assessment	HMD	L0-5	NA	NA	NA (NA)

by remote human operators wearing AR/VR HMDs to immersively operate the vehicle. Our findings show that many interaction modalities are being explored separately, and thus we encourage the use of sensor and interaction fusion, that is, the combination of multiple interaction modalities that are activated depending on the current context of being in a highly automated vehicle (e.g., work/leisure trip, entertainment/work task, private/public transport, location, time of day). Furthermore, different mobility approaches (e.g., shared public

transports) should be evaluated in combination with acceptable interaction types. AR interfaces can assist to convey the success or failure of given interactions, and provide feedback, as is already being done with smart home appliances using conversational artificial intelligence (Lopatovska et al., 2019). Such augmentations need not solely be visual, but can be auditory as well, for example. In the areas of automotive context awareness, and customization, as well as the level of customization/personalization that should be available to the

Table 7. *Summary of User Studies in Passenger Experiences Application Area*

References	Topic	Displays used	Vehicle		Dependent measures	Participants (female)
			Automation	Data type		
Wang, Charissis, and Harisson (2017)	Entertainment	Monitor	L0-5	S	Rating	20 (NA)
Wang, Charissis, Lagoo, Campbell, and Harrison (2019)	Rear passenger AR HUD	HUD	L0-2	O+S	Collisions, Rating, TAM	50 (NA)
Detjen, Geisler, and Schneegass (2020)	Reduce stress after passive accident experiences	Monitor	L3-4	S	Rating	43 (14)
Arnedo et al. (2019)	Cross-vehicle multiplayer games	HMD	L3-4	O+S	TOR Time, PXI Questionnaire, Interview	12 (3)
Sakamura et al. (2018)	Boarding system for AV passengers	HMD, monitor	L5	S	Rating	20 (NA)

users, in combination with augmented reality, we believe that research is still early and should be expanded. Particularly, the type of interaction suitable for the current situation should be evaluated. For example, in shared public vehicles, large AR windshield displays in combination with speech interaction might not be suitable from a proxemics and privacy viewpoint, which might not be the case in private vehicles.

4.5 Passenger Experiences

There were five studies in the Passenger Experiences application area we reviewed. Most of the studies explored higher levels of vehicle automation (see Table 7), with particular emphasis on fully automated driving reported by one study. HMDs and monitors were commonly used, while a HUD setup was only utilized in one study. All studies collected qualitative data, and two studies additionally evaluated quantitative criteria, such as TOR times, and number of collisions. Regarding subjective data, the Technology Acceptance Model (TAM), the Player Experience Questionnaire (PXI), and self-ratings using custom questionnaires were used most extensively. Four studies were designed as within-subject studies, and one as between-subject study. The within-subject studies were carried out with 24 participants on average, with almost one third of the subjects being female. Regarding the location, one of the studies took

place in an outdoor environment, the remaining studies were run in indoor lab environments.

4.5.1 Discussion. Increasingly, in-vehicle passenger activities and experiences are researched focussing on AR technology. Entertainment for all passengers can be achieved with AR interfaces. For example, as with head-up or windshield displays, side windows can also be utilized as see-through AR interfaces to display content for side and rear-seat passengers. Wang, Charissis, and Harisson (2017) explored visualizing entertainment and games on the side windows in order for younger passengers to stay entertained on longer rides. Wang, Charissis, Lagoo, Campbell, and Harrison (2019) investigated rear-seat commuters' preferences for activities in AR, and found that AR HUDs for rear-seat users can result in increased road safety by reducing distractions (Wang, Charissis et al., 2019). Stress and anxiety, and how they can be mitigated using appropriate interfaces, also play an important role in the context of passenger experiences (Detjen, Geisler, and Schneegass, 2020). AR has further been explored to guide passengers inside autonomous vehicles (Paredes et al., 2018). As the need for human vehicle operators/drivers will diminish in future automated driving scenarios, comprehensible guidance of passengers, notably in public transport, needs to be explored (Sakamura et al., 2018). Entertainment activities that involve multiple vehicles could also be a

Table 8. *Summary of User Studies in External HMI and Vulnerable Road Users Application Area*

References	Topic	Displays used	Vehicle		Dependent measures	Participants (female)
			Automation	Data type		
Chang, Toda, Sakamoto, and Igarashi (2017)	Interface design for communication between pedestrians and AVs	HMD	L5	O+S	Error/Accuracy, Time, Rating, Interview	15 (8)
Colley, Häkklä, Forsman, Pflöging, and Alt (2018)	Exterior surface displays	HUD	L0-5	S	Interview	8 (4)
Perez, Hasan, Shen, and Yang (2019)	Crossing behavior using AR	HMD	L0-5	O	Time, Distance	5 (NA)
Nguyen, Holländer, Hoggemüller, Parker, and Tomitsch (2019)	Projection-based communication between pedestrians and AVs	HMD	L5	S	Rating, Interview	18 (11)
Tong and Jia (2019)	Warning interface for pedestrians	HUD	L0-2	S	Rating	9 (3)
Maruhn, Dietrich, Prasch, and Schneider (2020)	Pedestrian behavior in AR	HMD	L5	O+S	Time, Perceived Safety, Rating	30 (15)

potential future AR experience, as Arnedo et al. (2019) explored.

Gamification concepts that incorporate real-world environments in conjunction with digital AR overlays in vehicle environments could represent a new approach to social driving experiences. As vehicle automation increases, and the driver becomes a passenger, this research will be particularly valuable as vehicles can be configured as mobile entertainment or work platforms. We thus encourage automotive UI researchers to consider comfort, anxiety, entertainment and work aspects using augmented reality technology for automated vehicles.

4.6 eHMI and Vulnerable Road Users

We examined a total of six studies in the eHMI and Vulnerable Road Users application area. We found that research was conducted almost entirely for fully automated vehicles (see Table 8). All studies utilized HMDs or HUDs for their eHMI concepts. Further, all but one study collected qualitative data. Additionally, quantitative data was reported by two studies. Quantitative

criteria included time and distance measurements, and error/accuracy, among others. Qualitative data included self-ratings and interviews. Five studies were designed as within-subject studies, and were conducted with 12 participants on average, with almost half of the subjects being female. With respect to the location, two studies were conducted outdoors, the remaining studies took place in indoor environments.

4.6.1 Discussion. External car bodies have the potential to provide an additional design space to a multitude of users. Currently, since the technical deployment of eHMIs for real vehicles is tedious and expensive, many research approaches prototype eHMIs concepts using virtual reality technology. To this end, information is displayed via AR interfaces to users, including pedestrians, cyclists, and other road users (Tong & Jia, 2019). For example, eHMIs can provide additional information, such as the driving state and intent, to other road users. Tong and Jia (2019) investigate a communication interface between an automated vehicle and pedestrians by visualizing warnings. Furthermore, Chang, Toda,

Sakamoto, and Igarashi (2017), Colley, Häkklä, Forsman, Pfleging, and Alt (2018), and Nguyen, Holländer, Hoggemueller, Parker, and Tomitsch (2019) explore several eHMI concepts, such as public displays, warning systems, and malfunction alerts. External HMIs are not constrained to display read-only information; however, they can be utilized for interaction purposes, such as route navigation for pedestrians (Asha, Anzum, Finn, Sharlin, & Sousa, 2020; Colley et al., 2018). We encourage researchers to investigate eHMIs not only for pedestrians, but also to consider other road users, including cyclists and drivers/passengers of other vehicles. The goal is to facilitate safe interactions and enhance communication between automated vehicles and vulnerable road users by establishing a common understanding between them. Additionally, the potential of vulnerable road users wearing AR glasses in future scenarios should be investigated further. For example, presenting visual cues on pedestrians' AR glasses through communicating intents between the road users.

4.7 Trust and Acceptance

In the Trust and Acceptance application area, a total of nine studies were reported. Our findings show that fully automated vehicles were most prominently researched (see Table 9); however, manual driving also received attention in the area of driver's attitude and perception towards AR in vehicles. Moreover, authors made use of all AR/VR devices, including HUDs, HMDs, CAVEs, and monitor setups in their study designs. Eight studies were designed to be within-subjects, and one study was designed as between studies. On average, 26 participants joined a within-subject study, with more than one-third of the subjects being female. All studies were conducted in indoor environments. All studies reported subjective data, and one-third of the studies included objective data in addition. Regarding objective data, heart rate variability (HRV), and gaze metrics were reported, while subjective feedback focussed on Trust Scale (TS), Trust in Automation Questionnaire (JB), Technology Acceptance Model (TAM), User Experience Questionnaire (UEQ), NASA TLX, and self-ratings.

4.7.1 Discussion. As the level of vehicle automation increased, more and more research efforts are invested into trust and acceptance in AD systems. For example, Ha, Kim, Seo, and Lee (2020) explore the effect of feedback types and perceived risk on trust in automated vehicles. Simple feedback design, such as descriptions of the vehicle's tasks, were found to increase trust in AVs. However, providing an abundance of explanation led to potential cognitive overload, and did not improve driver's trust levels towards the AV (Ha et al., 2020).

Wintersberger, Nicklas, Martlbauer, Hammer, and Riener (2020) found that the amount of feedback that should be provided to users is highly individual to users, and will require adaptive UIs to calibrate trust for future drivers/passengers. Wu et al. (2020) evaluate the acceptance of automated vehicles and found that AR cues had a positive impact on drivers by providing advice for their decision-making and reducing their anxiety and stress. Sawitzky, Wintersberger, Riener, and Gabbard (2019) explore AR traffic augmentation concepts to let drivers/passengers anticipate upcoming system decisions and maneuvers. Further, trust and acceptance should not only be evaluated from the driver's view, but also external users should be considered. For example, Colley et al. (2020) investigate trust in AVs by detecting and recognizing pedestrians' intent in order to calibrate driver's trust. The research of trust and acceptance encompasses HCI design (in-vehicle or external), feedback design (explainable artificial intelligence, i.e., using AR cues), and multiple users (i.e., driver, passengers, and vulnerable road users, including pedestrians). We identify challenges in this domain when it comes to determining trust over longer periods of time in longitudinal user studies, and the impact of reduced trust, eventually leading to distrust, and then subsequently aiming to regain trust in automated vehicles. Consequently, overtrust in AVs must be researched as well, such that drivers of conditionally automated vehicles stay in the loop, and alert, which could be facilitated using AR interfaces. Particularly, the level of system feedback could be calibrated (Filip, Meng, Burnett, & Harvey, 2016). In this regard, context awareness that encompasses the driver's physiological measures (e.g., stress), could be

Table 9. *Summary of User Studies in Trust and Acceptance Application Area*

References	Topic	Displays used	Vehicle		Dependent measures	Participants (female)
			Automation	Data type		
Häuslschmid, Bülow, Pfleging, and Butz (2017)	Trust using avatar, WIM, and indicators	Monitor	L5	S	UEQ, Rating, Interview	30 (12)
Wintersberger, Sawitzky, Frison, and Riener (2017)	Traffic augmentation	CAVE	L5	O+S	HRV, TS, TAM, Interview	26 (11)
Wintersberger, Frison, Riener, and Sawitzky (2019)	Evaluating potential of AR on trust	CAVE	L5	S	UX Curve, TS, TAM, Interview	18 (6)
Sawitzky, Wintersberger, Riener, and Gabbard (2019)	Traffic augmentation	HMD	L5	S	TAM, TS, JB, Interview	30 (10)
Oliveira, Burns, Luton, Iyer, and Birrell (2020)	System Transparency	HUD	L5	O+S	Trust Scale, Advanced Transport Telematics Survey, Rating, Interview	25 (4)
Wintersberger, Nicklas, Martlbauer, Hammer, and Riener (2020)	Explainable Automation, Adaptive UI	Monitor	L5	S	Rating, Interview	56 (37)
Ha, Kim, Seo, and Lee (2020)	Explanation types and perceived risk on trust	HMD	L5	S	Rating	51 (18)
Colley et al. (2020)	Recognizing pedestrians' intent	HMD	L5	O+S	NASA TLX, Trust in Automation Questionnaire, Rating, View Angle	15 (4)
Wu et al. (2020)	AR interface - driver's attitude and perception	Monitor	L0-2	S	Rating	16 (4)

considered to adjust the amount and transparency of system feedback.

4.8 Simulator and Motion Sickness

A total of five studies were reported in the Simulator and Motion Sickness application area. Our review reveals that research focussed on visual cues to tackle simulator sickness (see Table 10). Regarding vehicle automation, manual, conditional, and fully automated driving scenarios were explored. Further, most user studies reported using HUDs, while two user studies

utilized an HMD and monitor setup, respectively. Similarly, an analysis of the collected data types shows that objective criteria (2) and mixed data types (2) were examined most commonly, followed by subjective-only (1). Regarding objective data, driving times, error/accuracy, and acceleration/braking metrics were reported, while subjective feedback included SSQ, Presence Questionnaire (PQ), and the Misery Scale. All studies reported using a within-subject design. On average, 18 participants joined a within-subject study, with 38% of the subjects being female. Moreover, three studies were carried out in outdoor environments, and two studies took place in indoor lab settings.

Table 10. *Summary of User Studies in Simulator and Motion Sickness Application Area*

References	Topic	Displays used	Vehicle		Dependent measures	Participants (female)
			Automation	Data type		
Karl, Berg, Ruger, and Farber (2013)	Simulator sickness while driving	HMD	L0-2	O+S	Speed, Acceleration/Braking, SSQ	44 (15)
Okumura and Shinohara (2016)	Attention and fatigue	HUD	L0-5	O	Error/Accuracy	14 (7)
Kuiper, Bos, and Diels (2018)	Display positioning	HUD	L3	O+S	Time, Error/Accuracy, Misery Scale	18 (10)
Nakajima, Ino, Yamashita, Sato, and Kimura (2009)	MR sickness auditory stimuli	HUD	L0-2	O	Postural stability	7 (1)
Benz, Riedl, and Chuang (2019)	Comparison projector vs HMD	HMD, Monitor	L0-2	S	SSQ, Presence Questionnaire	60 (23)

4.8.1 Discussion. Simulator and motion sickness should be considered when running user studies in mixed reality environments in general, and on automated driving scenarios in particular. Simulator sickness is visually induced motion sickness (Rangelova & Andre, 2019), which can take place when stationary users move, or when head movements and the corresponding visual (virtual) display do not match Hettinger and Riccio (1992). Motion sickness can lead to nausea, disorientation, headache, and general discomfort, among others (Kennedy & Fowlkes, 1992). Kuiper, Bos, and Diels (2018) found that in-vehicle display positioning has an effect on motion sickness. Their findings show that utilizing a display at windshield height had the potential to mitigate motion sickness compared to a head-down display positioned at glove compartment height (Kuiper et al., 2018). Okumura and Shinohara (2016) achieved wider field of attention and lower eye-fatigue using a continuous-depth AR interface, in contrast to screen-fixed content positions. A wide range of proposed solutions to simulator/motion sickness are being explored, not just from the perspective of visual AR. It should be noted that subjective assessments (Simulator Sickness Questionnaire—SSQ, Presence Questionnaire—PQ) should be evaluated and reported for user studies, and physiological measures (e.g., heart rate variability, galvanic skin response) could further help to identify the underlying causes of simulator/motion sickness in au-

tomated driving (Classen, Bewernitz, & Shechtman, 2011). In addition, it is encouraged to address factors such as study duration and visual fidelity in VR simulations that influence the degree of immersion and presence (Bowman & McMahan, 2007). We encourage researchers to further investigate approaches on the mitigation of motion sickness in (conditionally) automated vehicles using world-fixed AR interfaces, as well as visual transitions between multiple displays in order to facilitate in-vehicle experiences.

5 Conclusion

5.1 Overall Summary

In this paper, we reported on the state of the art of AR technology used in human-computer interaction research in driving automation, spanning over the years 2009 to 2020. Following the PRISMA approach, we systematically reviewed papers from a wide range of journals and conferences, including 156 papers which reported user studies. We applied a structured method to outline the research domain by choosing relevant papers, and reviewing them in a standardized manner. The growing number of papers over the years 2009–2020 reveals that increasing levels of vehicle automation led to an increased usage of AR applications, accordingly.

Many different aspects of AR utilization in driving automation are being researched, showing that researchers in the community investigate topics on developing and improving human-machine interfaces for automated vehicles, as well as different perspectives, such as drivers, passengers, pedestrians, and other road users. Ideally, the present work provides researchers and practitioners with an overview of the current landscape, when they plan to engage in research and development of automated driving. Thus, one can extract information about the main research areas and emerging trends that have not been explored comprehensively so far.

5.2 Findings and Suggestions

Through our literature review, we found research gaps, and summarized these results into what we identify as the current high-level challenges in this domain.

This review identifies avenues for improved user study design, expanded use of field/outdoor studies, and a wider variety of evaluation methods. We found that user studies are overwhelmingly conducted with young, male, and technologically savvy participants, indicating the potential need for more inclusive and diverse participant sampling. We also note that AR often uses within-subject designs, which require smaller numbers of subjects compared to between-subject designs to achieve reasonable statistical significance. The median number of participants (at least 20 participants) seems adequate for all application areas. Overall, formal, laboratory-based experiments dominate AR studies; however, automotive AR researchers and practitioners are encouraged to do more studies in real-world outdoor use scenarios because the social and environmental impacts of outdoor locations may deviate from those of indoor locations. Therefore, the results of indoor studies cannot be directly extrapolated to outdoor sites.

This literature review is designed to give researchers and practitioners recommendations on methodological approaches (i.e., data collection methods and specific parameters) that can be followed when conducting AR user studies for automated driving research. Finally, we present a set of suggestions that should be addressed in

future studies that use augmented reality technology in the automotive domain:

- **User study design and reporting.** As this literature review highlights, many user studies are conducted with a majority of young, male subjects from the university population. We also found that reports on women and other genders are often left out, suggesting a lack of inclusivity in this area of research. In addition, any user study that does not include information on the participant population, demographics, study design, or experimental tasks is hard to replicate, and its results are virtually impossible to generalize. Finally, we also suggest specifying the keywords of the paper according to the study design, that is, if a virtual reality driving simulator was used, we recommend adding this information in the keywords or in the abstract section. This is a general requirement to improve the quality of user study reporting and to guide researchers in the field on how to conduct good mixed reality user studies.
- **Heuristics, benchmarking, and data sets.** Establishing a common set of criteria for evaluating HMI designs (e.g., structural metrics for user interfaces) should be pursued to automate design evaluations and reduce reliance on lengthy user studies. In addition, datasets sharing should be undertaken to facilitate comparisons between studies and to improve these criteria.
- **Additional focus on user experience.** With regard to specific research questions, we advocate investigating constructs such as personalization and well-being in addition to the usual research topics such as safety, trust & acceptance, and situational awareness. Moreover, user-centered design of vehicle features and a focus on user experience could help to promote public acceptance of automated vehicles (Frison, Wintersberger, Riener, & Schartmüller, 2017).
- **Fidelity and realism.** As technological advances in AR devices such as field of view, resolution, and processing power continue to improve, we propose to visually enhance virtual and mixed reality

environments and create digital twins of parts of the real world to convey a higher degree of realism.

- **Measuring simulator sickness.** When running VR driving simulator experiments to emulate AR content, we recommend using standardized questionnaires (e.g., SSQ) to evaluate the simulator sickness of study participants before and after the experiment. We noticed that many researchers use such questionnaires at the end of the experiment; however, we suggest that they should also be used before the VR simulation to analyze its effects on participants' nausea, disorientation, etc.

Our analysis highlights potential for investigating the use of augmented reality in higher levels of vehicle automation (SAE Level 3 and above). Finally, we suggest a number of research topics that should be addressed in future experiments on augmented reality in automated driving:

- **Scenarios and analysis for mixed traffic.** Scenarios and analysis for mixed/mixed traffic. Since we still have decades of shared manual and automated vehicles on the road ahead of us, we advocate further investigation of mixed traffic situations from a human-machine interface perspective. Both internal and external AR interfaces in combination cooperative and teleoperated driving should therefore be explored. For conditionally automated vehicles, the design of take-over requests, and context-aware use cases to transition between manual and automated driving, should be further researched.
- **Passenger experiences, and user-centered designs and interactions.** As active drivers become passive passengers with increasing vehicle automation, work and entertainment activities, such as interactions with in-vehicle infotainment systems with an AR interface, and collaborations among passengers, should be further explored. The effect of simulator and motion sickness and their role in human-machine interface design should also be studied. Since AR information can be "placed" in the external environment, world-fixated applications should be given more research attention. Furthermore, per-

sonalization and customization concepts should be examined according to the individual preferences of users/drivers. In addition, issues such as context awareness and multimodal interaction require further research attention for higher levels of vehicle automation.

- **Trust and acceptance of AVs.** The design of feedback and explainable artificial intelligence, for example, displaying AR cues as to what the vehicle "sees," including vulnerable road users, should be explored, as should the development of trust in AVs and how overconfidence can be addressed with AR interfaces. In conditionally automated driving, when the vehicle takes over the driving task, the involvement of pedestrians, cyclists, and other vulnerable road users must be appropriately studied. Trust and acceptance must be measured not only from the driver's perspective, but also in consideration of other road users.
- **Visual complexity analysis for automotive AR applications.** In line with the desktop and mobile domain, for example, Riegler and Holzmann (2018) and Zen and Vanderdonckt (2014), we encourage investigating the quantification of visual complexity of automotive AR user interfaces, both structural (static), and behavioral (dynamic) for different levels of vehicle automation, to support rapid prototyping of AR head-up and windshield display interfaces, and the development and usage of automotive AR apps in the future. Adaptive UIs could further be researched in terms of calibrating trust in AVs from the viewpoint of drivers, passengers, and other road users.
- **Outdoor studies with AR, and mitigating motion sickness.** Currently, most studies of AR in automated driving are carried out indoors due to safety and technology concerns. With the advances in AR and vehicle automation technology, we propose to progressively run mixed reality studies in real vehicles on outdoor test tracks. As user studies may shift from driving simulators to real vehicles, mitigating motion sickness by utilizing AR interfaces should receive more research focus.

5.3 Final Thoughts and Future Plans

The overall goal of this literature review was to provide a broad overview of augmented reality usage and applications for vehicles with varying levels of automation over the past decade. This work is envisioned to serve as inspiration for researchers and practitioners in specific automotive augmented reality application areas as they formulate their own research agendas. We also sought to chart the historical evolution from low-immersion monitor/HUD setups to more advanced CAVE/HMD setups and to simulate continuous-depth AR content in VR environments. We also summarized methodological approaches, highlighted their limitations, and provided recommendations for future research. Future plans include examining each application area in more detail (e.g., augmented reality for takeover requests for non-driving related tasks) and creating more detailed and targeted overviews. There is also a need to keep this analysis up to date with each completed release year, starting in 2021. AR in automated driving is still an evolving research field that needs to be continuously monitored. We see interest and efforts in this area increasing, and we look forward to continued work.

Conflict of Interest Statement

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Author Contributions

All authors contributed to the entire review process and the manuscript. AR1 initiated the process with database search, data collection, and analysis. AR1, AR2, and CH all reviewed and collected data. AR1 took the lead in writing the manuscript.

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